Directional Alarm Clock

Levi Terry

University of Advancing Technology

Robotics Engineering

Digital Maker and Fabrication, SIP311

Problem Statement

When normal alarm clocks go off, they can easily wake up everyone in a room. This can become a problem if multiple people share a bedroom and have different wake up schedules. This is especially common in college dorm rooms, for example. My product will solve this by targeting the alarm to a specific person, so not everyone in the room will wake up if they don't need/want to.

This alarm clock will use an ultrasound speaker for directional audio to target a specific person, so only the person using the alarm will be awoken. Along with directional audio, the clock will require the completion of a task to disable the alarm clock to activate the brain and get the user ready for the day.

For Statistics see Appendix A: Statistics

Innovation Claim

The Directional Alarm Clock is a user-specific method of alerting users with the use of an ultrasound speaker to direct the alarm towards the specific sleeper who needs to be woken up without disturbing other nearby sleepers.

Technical Expertise & Skills

This project requires experience in the following fields:

- Programming
- Electronic design
- 3D modeling
- Fabrication

Project Description & Scope

This project combines an alarm clock and an ultrasonic speaker creating a directional alarm clock that will be able to "target" one specific person as to not disturb others that may want to continue sleeping.

I'm not planning on working with a team for this project with a team, but I am open to collaboration and help from other people.

As for timeline I'm planning on making progress throughout this and next semester to create a MVP consisting of an ESP32-powered touchscreen for setting alarms and an ultrasonic directional speaker.

Target Market

This will be primarily targeted for people who sleep in the same ro with someone else with each person having a different schedule. This might include:

- Married Couples
- Roommates
- Siblings

Prior Art

- Focusonics® Model B - Focusonics - Directional Speaker
- CLOCKY Extra Loud Alarm Clock - Amazon - Alarm clock that tries to run away
- Sonic Bomb Alarm Clock with Super Shaker - Sonic Alert - Loud alarm clock with function to shake bed

Appendix A: Statistics

Projected value of furniture market (Furniture, Furnishings & Household Items. n.d.)

Morning Feelings

Approximately 53.86% of people feel dreadful when their alarm goes off in the morning. Additionally, 27.12% of people feel anxious, 11.24% feel energized, 2.32% feel excited, and 0.32% of people feel awake. (Morgan 2019)

Snooze Usage

Most people hit the snooze button at least once. 35.57% of women and 43.39% of men never press snooze. While 6.31% of women and 5.65% of men say they press snooze over 3 times.

Additionally, 61.94% only have 1 alarm set, 21.16% set 2, 12.44% set more than 2, and 4.46% of people don't set alarms. (Morgan 2019)

References

Furniture, Furnishings & Household Items. (n.d.). Statista. https://www.statista.com/markets/415/topic/469/furniture-furnishings-household-items/#statistic1

Morgan, B. (2019, May 3). How Many Times Do Most People Hit the Snooze Button? (And More Fascinating Alarm Clock Stats). apartment therapy. https://www.apartmenttherapy.com/how-many-times-do-most-people-hit-the-snooze-button-and-more-fascinating-alarm-clock-stats-243011

Proof of Concept/Early-Stage Prototype

Insert image(s), sketch, schematic, flowchart, rendering, etc.

Q&A

Levi Terry, Robotics and Embedded Systems + Digital Maker and Fabrication

This alarm clock will use an ultrasound speaker for directional audio to target a specific person, so only the person using the alarm will be awoken. Along with directional audio, the clock will require the completion of a task to disable the alarm clock to activate the brain and get the user ready for the day.